

SAFETY DATA SHEET

(SDS)

This safety data sheet complies with the requirements of:
Regulation (EC) No. 1907/2006 and Regulation (EC) No. 1272/2008, (EU) No. 453/2010

Revision Date 10-Feb-2016

WAI2 - EGHS - EUROPEAN

Revision Number 2

SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product Identifier

Product Name Iron 1
Product No AC4078-ACT
Pure substance/mixture Mixture

Contains Thioglycolic Acid

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Use as laboratory reagent
Uses advised against No Information available

1.3. Details of the supplier of the safety data sheet

Manufacturer, Importer, Supplier Thermo Orion Inc. (Part of Thermo Fisher Scientific, Inc.)
Water Analysis Instruments
22 Alpha Road
Chelmsford, MA 01824, USA
1-978-232-6000

E-mail address wai.techservbev@thermofisher.com

Made in USA

1.4. Emergency telephone number 24 Hour Emergency Phone Number
CHEMTREC®
Within USA and Canada: 1-800-424-9300
Outside USA and Canada: 1-703-527-3887
(collect calls accepted)

SECTION 2. HAZARDS IDENTIFICATION**2.1. Classification of the substance or mixture**

Classification - Mixture

Classification according to Regulation (EC) No. 1272/2008 [CLP]

Acute Toxicity - Oral	Category 3 - (H301)
Acute toxicity - Dermal	Category 4 - (H312) - (H330)
Acute toxicity - Inhalation (Vapors)	Category 1 - (H330) - (H331)
Skin corrosion/irritation	Category 1 Sub-category B - (H314)
Serious eye damage/eye irritation	Category 1 - (H318)

2.2. Label elements**Product Identifier**

Contains Thioglycolic Acid

**Signal Word**

Danger

Hazard Statements

H301 - Toxic if swallowed

H314 - Causes severe skin burns and eye damage

H312 - Harmful in contact with skin

H330 - Fatal if inhaled

Precautionary Statements

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor/physician

P322 - Specific measures (see .? on this label)

P304 + P340 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

P320 - Specific treatment is urgent (see .? on this label)

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

P321 - Specific treatment (see supplemental first aid instructions on this label)

P303 + P361 + P353 - IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower

P280 - Wear eye protection/ face protection

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor/physician

P202 - Do not handle until all safety precautions have been read and understood

2.3. Other hazards

No information available

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**3.1. Substances**

Component	Chemical Formula	EC-No.	CAS-No	Weight %	CLP Classification - Regulation (EC) No 1272/2008	REACH Reg. No
Ammonium thioglycolate	No information available	EEC No. 226-540-9	5421-46-5	40 - 50%		No information available
Thioglycolic Acid	No information available	EEC No. 200-677-4	68-11-1	30 - 40%	Acute Tox. 3 (H301) Acute Tox. 3 (H311) Acute Tox. 3 (H331) Skin Corr. 1B (H314)	No information available
Water	No information available	EEC No. 231-791-2	7732-18-5	20 - 30%		No information available

Note *The exact percentage (concentration) of composition has been withheld as a trade secret

Full text of H- and EUH-phrases: see section 16

SECTION 4: FIRST AID MEASURES**4.1. Description of first aid measures**

General Advice	Use first aid treatment according to the nature of the injury. For further assistance, contact your local Poison Control Center. Show this safety data sheet to the doctor in attendance.
Eye Contact	In case of eye contact, remove contact lens and rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Obtain medical attention.
Skin Contact	Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. If symptoms persist, call a physician.
Inhalation	Move to fresh air. If breathing is difficult, give oxygen. Get medical attention if symptoms occur.
Ingestion	Clean mouth with water and drink afterwards plenty of water. Do not induce vomiting. Call a physician or Poison Control Center immediately.
Protection of First-aiders	Use personal protective equipment. See section 8 for more information. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

4.2. Most important symptoms and effects, both acute and delayed

Most important symptoms/effects No information available

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable Extinguishing Media

No information available

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Personal Precautions

Use personal protective equipment. Evacuate personnel to safe areas.

6.2. Environmental precautions

Environmental Precautions

Beware of vapors accumulating to form explosive concentrations. Vapors can accumulate in low areas.

6.3. Methods and material for containment and cleaning up

Methods for Containment

Prevent further leakage or spillage if safe to do so.

Methods for Cleaning Up

Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.

Reference to Other Sections

Refer to protective measures listed in Sections 7 and 8

See Section 8 for information on appropriate personal protective equipment

See Section 12 for additional Ecological Information

See Section 13 for additional waste treatment information

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Advice on safe handling

To avoid risks to human health and the environment, comply with the instructions for use. Wear personal protective equipment. Avoid breathing dust/fume/gas/mist/vapors/spray. Ensure adequate ventilation, especially in confined areas.

General hygiene considerations

Handle in accordance with good industrial hygiene and safety practice.

7.2. Conditions for safe storage, including any incompatibilities

Storage Conditions

Keep container tightly closed in a dry and well-ventilated place. Store at room temperature in the original container. Keep away from direct sunlight.

7.3. Specific end use(s)

Specific Use(s)

Use as laboratory reagent

Risk Management Methods (RMM)

The information required is contained in this Safety Data Sheet.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**8.1. Control parameters**

Component	European Union	The United Kingdom	France	Spain	Germany
Thioglycolic Acid 68-11-1	-	STEL: 3 ppm 15 min STEL: 11.4 mg/m ³ 15 min TWA: 1 ppm 8 hr TWA: 3.8 mg/m ³ 8 hr	TWA / VME: 1 ppm (8 heures). TWA / VME: 5 mg/m ³ (8 heures). Peau	TWA / VLA-ED: 1 ppm (8 horas) TWA / VLA-ED: 3.8 mg/m ³ (8 horas) Piel	Haut
Component	Italy	Portugal	The Netherlands	Finland	Denmark
Thioglycolic Acid 68-11-1	-	TWA: 1 ppm 8 horas Pele	-	TWA: 1 ppm 8 tunteina TWA: 3.8 mg/m ³ 8 tunteina STEL: 3 ppm 15 minuutteina STEL: 11 mg/m ³ 15 minuutteina Iho	TWA: 1 ppm 8 timer TWA: 5 mg/m ³ 8 timer
Component	Austria	Switzerland	Poland	Norway	Ireland
Thioglycolic Acid 68-11-1	Haut MAK-KZW: 2 ppm 15 Minuten MAK-KZW: 8 mg/m ³ 15 Minuten MAK-TMW: 1 ppm 8 Stunden MAK-TMW: 4 mg/m ³ 8 Stunden	Haut/Peau STEL: 2 ppm 15 Minuten STEL: 8 mg/m ³ 15 Minuten TWA: 1 ppm 8 Stunden TWA: 4 mg/m ³ 8 Stunden	STEL: 8 mg/m ³ 15 minutach TWA: 4 mg/m ³ 8 godzinach	TWA: 1 ppm 8 timer TWA: 5 mg/m ³ 8 timer STEL: 1 ppm 15 minutter. STEL: 5 mg/m ³ 15 minutter.	TWA: 1 ppm 8 hr. TWA: 5 mg/m ³ 8 hr. STEL: 3 ppm 15 min STEL: 15 mg/m ³ 15 min

Derived No Effect Level (DNEL) No information available**Predicted No Effect Concentration (PNEC)** No information available**8.2. Exposure controls**

Engineering Measures

Showers
Eyewash stations
Ventilation systems

Personal protective equipment

Eye/face Protection Wear chemical splash goggles and face shield. If splashes are likely to occur, wear: Goggles.

Skin and body protection Wear protective gloves/clothing.

Respiratory Protection No protective equipment is needed under normal use conditions. In case of inadequate ventilation wear respiratory protection.

Environmental exposure controls No information available**SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES****9.1. Information on basic physical and chemical properties**

Physical State Liquid

Appearance Colorless to pale pink

Odor	Sharp and characteristic
Odor Threshold	No information available
PH Range	2.5-6.5

<u>Property</u>	<u>Values</u>	<u>Remarks • Method</u>
Melting point/freezing point	No information available	
Boiling Point/Range	96 °C / 204.8 °F	
Flash Point (High in °C)	125 °C / 257 °F	
Evaporation Rate	No information available	
Flammability (solid, gas)	No information available	
Flammability Limit in Air		
Upper flammability limit:	No information available	
Lower flammability limit:	No information available	
Vapor pressure	No information available	
Vapor Density	No information available	
Specific Gravity	No information available	
Water Solubility	Soluble in water	
Solubility in other solvents	No information available	
Partition coefficient	No information available	
Autoignition Temperature		
Decomposition Temperature	No information available	
Kinematic viscosity	No information available	
Dynamic viscosity	No information available	
Explosive Properties	No information available	
Oxidizing Properties	No information available	

9.2. Other information

Softening Point	No information available
Molecular Weight	No information available
VOC Content(%)	No information available
Density	No Information available
Bulk Density	No information available

SECTION 10: STABILITY AND REACTIVITY**10.1. Reactivity**

No information available

10.2. Chemical stability

Stable under normal conditions

Explosion Data

Sensitivity to Mechanical Impact	None
Sensitivity to Static Discharge	None

10.3. Possibility of hazardous reactions

None under normal processing

10.4. Conditions to avoid

Extremes of temperature and direct sunlight

10.5. Incompatible materials

No information available

10.6. Hazardous decomposition products

Thermal decomposition can lead to release of irritating gases and vapors

SECTION 11: TOXICOLOGICAL INFORMATION**11.1. Information on toxicological effects**

Acute Toxicity**Product Information**

Product does not present an acute toxicity hazard based on known or supplied information.

Inhalation	No information available
Eye Contact	No information available
Skin Contact	No information available
Ingestion	No information available

Unknown Acute Toxicity 41 % of the mixture consists of ingredient(s) of unknown toxicity.

The following values are calculated based on chapter 3.1 of the GHS document

ATEmix (oral)	139.00 mg/kg
ATEmix (dermal)	1,614.00 mg/kg
ATEmix (inhalation-gas)	107.90 ppm
ATEmix (inhalation-dust/mist)	0.95 mg/L
ATEmix (inhalation-vapor)	0.41 mg/L

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Thioglycolic Acid	LD50 = 73 mg/kg (Rat)	LD50 = 848 mg/kg (Rabbit)	LC50 = 56.7 ppm (Rat) 4 h
Water	LD50 > 90 mL/kg (Rat)		

Skin Corrosion/Irritation	No information available
Serious eye damage/eye irritation	No information available
Sensitization	No information available
Mutagenic Effects	No information available
Carcinogenic effects	No information available
Reproductive Effects	No information available
STOT - single exposure	No information available
STOT - repeated exposure	No information available
Aspiration hazard	No information available

SECTION 12. ECOLOGICAL INFORMATION

12.1. Toxicity

72% of the mixture consists of components(s) of unknown hazards to the aquatic environment

12.2. Persistence and degradability

No information available

12.3. Bioaccumulative potential

No information available

12.4. Mobility in soil

No information available

12.5. Results of PBT and vPvB assessment

No information available

12.6. Other adverse effects

No information available

Endocrine Disruptor Information

No information available

SECTION 13. DISPOSAL CONSIDERATIONS**13.1. Waste treatment methods****Waste from Residues / Unused Products**

Disposal should be in accordance with applicable regional, national and local laws and regulations.

Contaminated Packaging

Improper disposal or reuse of this container may be dangerous and illegal.

SECTION 14: TRANSPORT INFORMATION**IMDG/IMO**

14.1 UN-No	1760
14.2 Proper Shipping Name	Corrosive Liquid, n.o.s. (contains ammonium thioglycolate)
14.3 Hazard Class	8
14.4 Packing Group	II
14.5 Marine Pollutant	No
14.6 Special Provisions	None
EmS No.	F-A, S-B
14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code	No information available

ICAO

14.1 UN-No	1760
14.2 Proper Shipping Name	Corrosive liquid, n.o.s.(contains ammonium thioglycolate)
14.3 Hazard Class	8
14.4 Packing Group	II
14.5 Environmental hazard	Not Applicable
14.6 Special Provisions	None

IATA

14.1 UN-No	1760
14.2 Proper Shipping Name	Corrosive Liquid, n.o.s. (contains ammonium thioglycolate)
14.3 Hazard Class	8
14.4 Packing Group	II
14.5 Environmental hazard	Not Applicable
14.6 Special Provisions	None
ERG Code	8L

SECTION 15: REGULATORY INFORMATION**15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture****European Union**

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents

at work

International Inventories

USINV	Complies
CANINV	Complies
EINECS/ELINCS	Complies
ENCS	Complies
IECSC	Complies
KECL	Does not Comply
PICCS	Complies
AICS	Complies

USINV/ TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

CANINV/ DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

AICS - Australian Inventory of Chemical Substances

15.2. Chemical safety assessment

A Chemical safety assessment according to regulation (EC) No. 1907/2006 is not required

SECTION 16: OTHER INFORMATION**Key or legend to abbreviations and acronyms used in the safety data sheet****Full text of H-Statements referred to under section 3**

H301 - Toxic if swallowed

H314 - Causes severe skin burns and eye damage

H312 - Harmful in contact with skin

H330 - Fatal if inhaled

Legend - SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

TWA	TWA (time-weighted average)	STEL	STEL (Short Term Exposure Limit)
Ceiling	Maximum limit value	*	Skin designation

Prepared By Environmental, Health and Safety

Prepared For Thermo Fisher Scientific Inc.

Issue Date No information available

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Reason for revision SDS sections updated.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006

Disclaimer

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End of Safety Data Sheet